

PROFILE

- Passionate about the convergence of Electronics and Computer Science, I am driven to merge technologies to everyone's life. My academic journey included hands-on projects and coursework focused on advanced system design, optimization, as well as machine learning and software development. Proficient in hardware-software integration and familiar with programming languages like C++ and Python. Eager to apply my knowledge and skills in a dynamic engineering role, contributing to innovative solutions in computer systems development.

EDUCATION

- **Ph.D. Computer Science, University of California - Riverside** Sep 2024 - Current
Riverside, California
- **B.S. Computer Engineering, University of California - San Diego** Sep 2019 - Sep 2023
San Diego, California

PROJECTS & CONTRIBUTIONS

- **RTSpMSpM: Harnessing Ray Tracing for Efficient Sparse Matrix Computations**
Developed and published a novel framework leveraging Nvidia RTX ray tracing hardware to accelerate sparse matrix computations. Demonstrated significant performance benefits over traditional SpGEMM methods using OptiX and CUDA.
Published Paper: ACM Digital Library: RTSpMSpM Publication
Lab: ESCAL, Riverside Sep 2023 - Present
- **Machine Learning in Music**
Projects include random music generation, classifying audio genre, autoencoder network... etc. Course: CSE 190
June 2023 - August 2023
- **Android Software Engineering**
Interactive map & social media application development. Course: CSE 110 January 2023 - March 2023

TECHNICAL SKILLS

- **Programming and Scripting Languages**
Python, C++, C, Bash
- **Libraries and Frameworks**
NVIDIA OptiX, CUDA, cuSPARSE, NumPy, Docker
- **Tools and Debuggers**
GitHub, gdb, jdb, vim
- **Operating Systems and Simulation Software**
Linux (Ubuntu), Windows, MATLAB, LTSpice, Multisim, Proteus